EXATHON INDIA 2026

TraceBack AI

Not just an alarm. The evidence behind it.

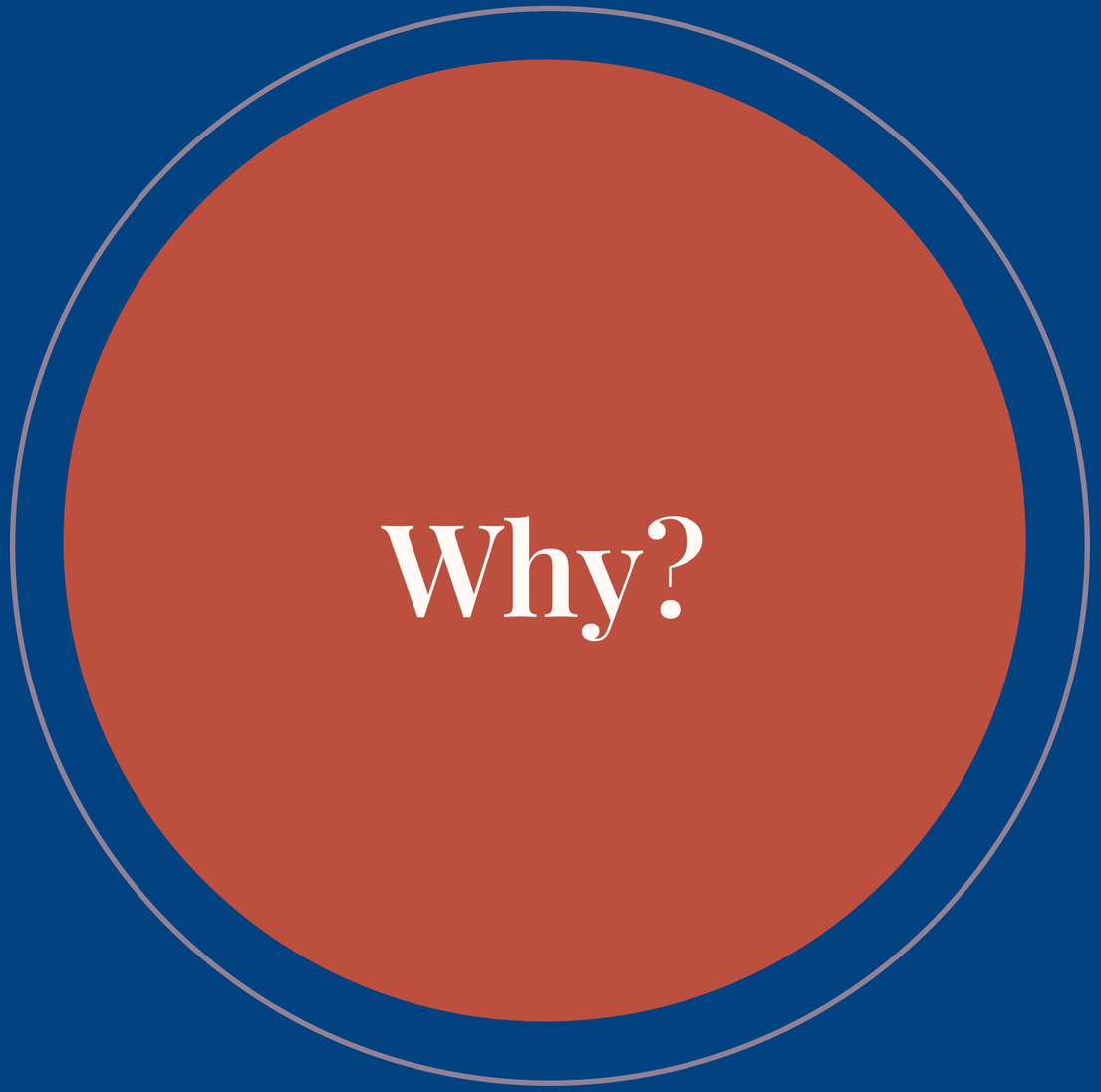
Created by Jyothik Bollineni and Vedhanth Krishnan

Factories, Fleets & Supply Chains · github.com/vedhanthkrishnan/TraceBack-AI

THE PROBLEM

“Machine 104: critical.”

An alarm says when. Engineers need to know why.



Three answers, with evidence

When did it start?

The first sustained drift, marked on the health trend.

Timeline

Which sensors?

Ranked by deviation, drift rate and persistence.

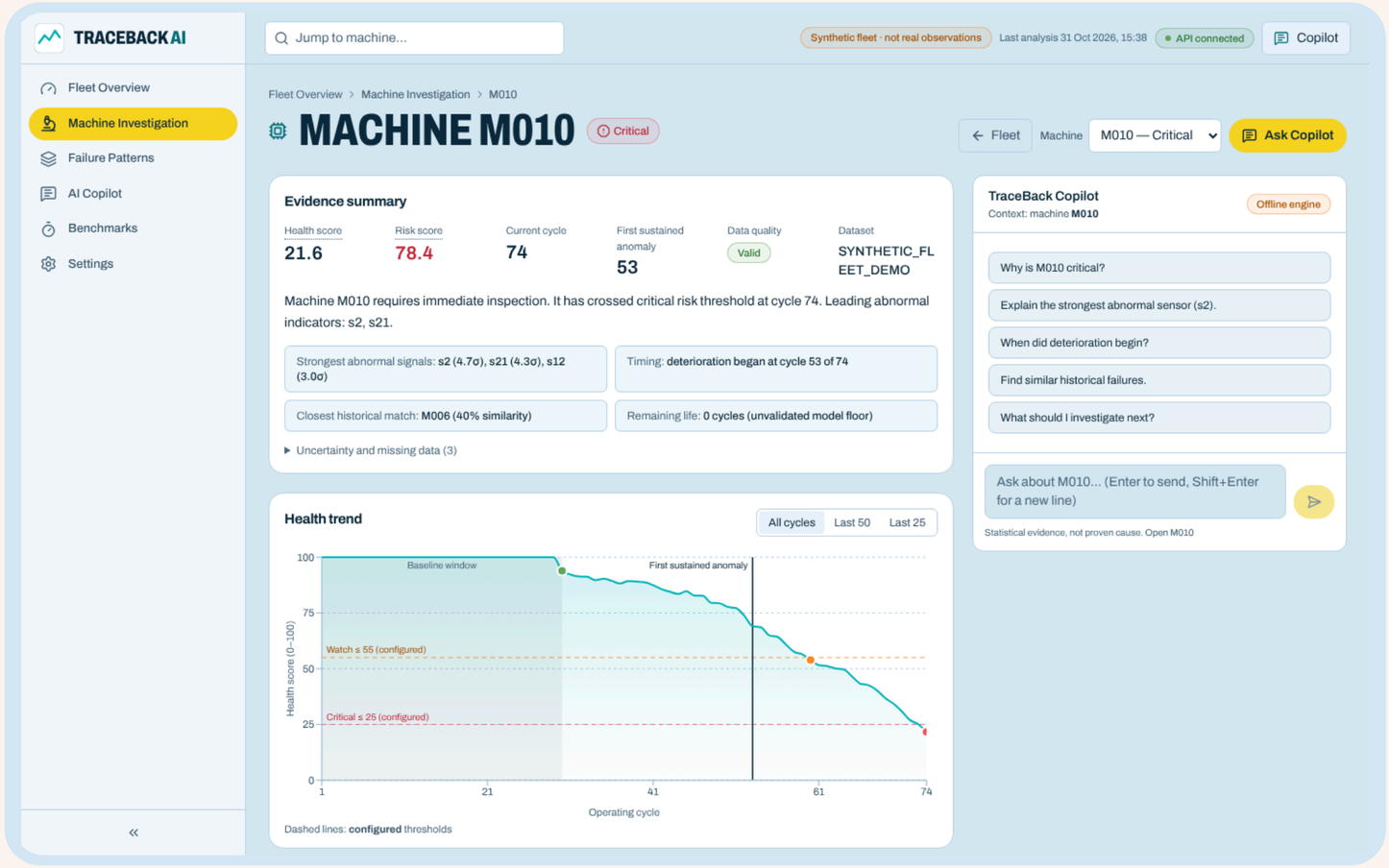
Root cause

Seen it before?

Past failures with a similar pattern, side by side.

Failure patterns

One screen per machine



Health trend

Where drift began

Ranked sensors

The evidence

Ask Copilot

Why it matters

Demo fleet, analysed

Live from the pipeline, on a synthetic fleet.

Machines

12

Analysed in one pipeline run

On watch

7

Early drift, inspect soon

Critical

2

Past the critical threshold

Top sensor, M010

4.7σ

Sensor s2 against its own healthy baseline

AI Copilot

Cited

Every answer links to its evidence.

GROUNDED ANSWERS

Every claim links back

Copilot reads the same evidence the dashboard shows.

Click a citation to jump to the exact sensor and cycle.

Missing data is stated, never guessed.

Runs offline by default. OpenAI or Gemini can be plugged in.

Three layers, one pipeline

Exasol SQL

Baselines, drift and risk scores, computed in SQL. Falls back to SQLite offline.

Analytics

FastAPI

One validated JSON API for fleet, evidence, matches and Copilot.

Service

React

Fleet overview, machine investigation, patterns and Copilot.

Interface

Synthetic machines

12

Demo data, not real observations.

THE HONEST PART

What it does not claim

Evidence, not proof. Statistics point to a cause; an inspection confirms it.

Remaining life is unvalidated. Shown as a model floor, not a promise.

Copilot is offline by default. A live LLM is optional.

Next: real fleet data and a per-cycle sensor history.

Thank you

Code, run guide and demo are on GitHub.

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Ask away